

Yibin Dong

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Institute of Frontier and Interdisciplinary Science, Shandong University, China

No.72 Binhai Road, Qingdao, Shandong Province, P.R. China

EDUCATION

Postdoctoral Researcher

Shandong University

2026 – Present

PhD in Condensed Matter Physics

Shandong University

2020 – 2025

PhD Thesis: Computational Simulation of Radiation Damage in Grain Boundaries and Phase Boundaries in Metal Materials

BSc in Physics

Shandong University

2015 – 2019

RESEARCH INTERESTS

- Atomistic simulations of defects, interfaces and their motions, interactions
- Computational simulation of plastic deformation and fracture mechanisms in materials;
- Multiscale computational simulation methods.

SELECTED PUBLICATIONS

- S. Zhang[#], **Y. Dong[#]**, Y. Sun, Y. Liu, L. Sun, H. Zhao, N. Gao, Z. Wang. "Improved Radiation Resistance in Metals via Adaptive Martensitic Transformation", **Nature Communications**, 2025, 16: 10995 (co-first author)
- **Y. Dong**, F. Wang, W. Setyawan, F. Gao, X. Wang, N. Gao. "Enhanced Resistance to Radiation and Magnetic Anisotropy Change by Periodic Pattern of Dislocation Networks at a W/Fe Interface", **Acta Materialia**, 2025. 288: 120880
- H. Chang[#], **Y. Dong[#]**, W. Setyawan, Y. Cui, M. Yu, W. Ma, L. Sun, X. Wang, N. Gao. "Effect of Radiation Defects on Grain Boundary Evolution under Shock Loading", **Journal of Materials Research and Technology**, 2024. 31: 698–707 (co-first author)

CONFERENCE PRESENTATIONS

- *Radiation Damage Mechanisms at W/Fe Interfaces*. International Conference on Fusion Reactor Materials-22 (ICFRM22), Shizuoka, Japan, September 2025. **[Poster]**
- *Study on the Influence of Periodic Dislocation Networks on Radiation Resistance of W/Fe Interface*. 9th China Fusion Reactor Materials Conference, Xi'an, China, August 2025. **[Oral]**
- *Improved Radiation Resistance in Metals via Adaptive Martensitic Transformation*. 12th Reactor Physics and Nuclear Materials Conference, Zhuhai, China, October 2025. **[Oral]**

TECHNICAL SKILLS

- Computational Methods: Molecular Dynamics (LAMMPS), Density Functional Theory (VASP), Ab initio Molecular Dynamics (CP2K);
- Simulation Areas: Radiation Damage, Fracture Mechanical, Shock Loading, Phase Transformations;
- Programming: Python, Fortran, MATLAB, Shell.